

Math706 – Numerical Linear Algebra

Midterm Exam

You must show all of your work to get a credit for a correct answer.

1. (15 points) Let $\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.
 - (a) Find a full SVD for $\mathbf{A}$.
 - (b) Compute $\|\mathbf{A}\|_1$, $\|\mathbf{A}\|_2$, $\|\mathbf{A}\|_\infty$, $\|\mathbf{A}\|_F$.
2. (20 points) Let $\mathbf{A} \in \mathbb{C}^{m \times m}$.
 - (a) Prove that $\mathbf{A}$ has a full QR factorization.
 - (b) Under what conditions is the QR factorization unique if $\mathbf{A}$ is full rank?
 - (c) Write out a pseudo code for the QR factorization using modified Gram-Schmidt orthogonalization.
3. (15 points) Suppose that $\mathbf{x}, \mathbf{y} \in \mathbb{C}^m$ are unit vectors, i.e., $\|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 = 1$.
 - (a) Construct a unitary matrix $\mathbf{F}$ such that $\mathbf{F}\mathbf{x} = \mathbf{y}$.
 - (b) What can be said for general $\mathbf{x}, \mathbf{y}$?
4. (15 points) Given $\mathbf{A} \in \mathbb{C}^{m \times n}$ of full rank n and $\mathbf{b} \in \mathbb{C}^m$, consider the block 2×2 system of equations
$$\begin{bmatrix} \mathbf{I} & \mathbf{A} \\ \mathbf{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{r} \\ \mathbf{x} \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix}.$$
Show that this system has a unique solution $(\mathbf{r}, \mathbf{x})^T$ and that the vectors $\mathbf{r}$ and $\mathbf{x}$ are the residual and the solution of the least squares problem $\min_{\mathbf{x} \in \mathbb{C}^n} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2$.
5. (20 points)
 - (a) Define the three terms "condition of a problem", "backwards stability of an algorithm" and "accuracy of an algorithm".
 - (b) Prove that a backwards stable algorithm applied to a problem with condition number κ is *accurate* and provide an error estimate in terms of κ .
6. (15 points) Let $\mathbf{A} \in \mathbb{C}^{m \times m}$. Suppose for each k with $1 \leq k \leq m$, the upper-left $k \times k$ block $\mathbf{A}_{1:k, 1:k}$ is nonsingular.
 - (a) Show that $\mathbf{A}$ has an LU factorization.
 - (b) What can you say about the sparsity of $\mathbf{L}$ and $\mathbf{U}$ if $\mathbf{A}$ is banded with bandwidth $2p + 1$, i.e., $a_{ij} = 0$ for $|i - j| > p$?

Extra Credit: (10 points)

Let $\mathbf{A} \in \mathbb{C}^{m \times n}$ has full rank n and $\mathbf{B} = \mathbf{A}(\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*$. Prove $\|\mathbf{B}\|_2 = 1$.